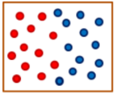
Ensemble Methods

Bagging

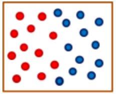
Bagging: Bootstrap Aggregation



- Bootstrap: Sampling the training data with replacement (N, <N)

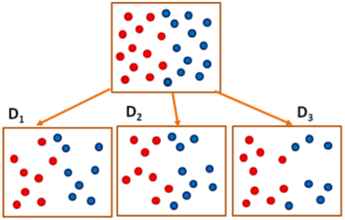
Bagging: Bootstrap Aggregation

- **Bootstrap:** Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times



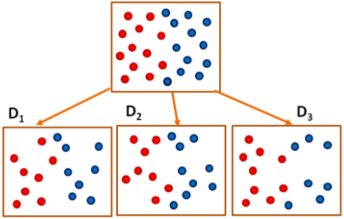
Bagging: Bootstrap Aggregation

- Bootstrap: Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times



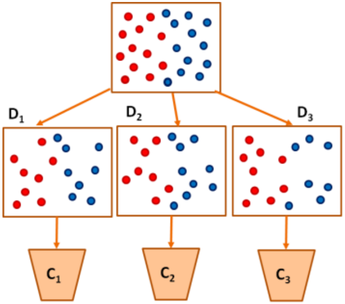
Bagging: Bootstrap Aggregation

- Bootstrap: Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times
  - Train a classifier, C_i on D_i.



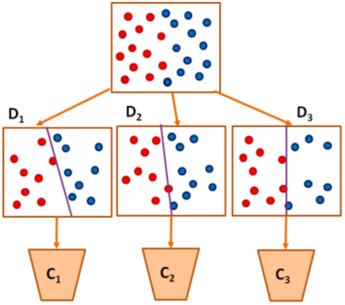
Bagging: Bootstrap Aggregation

- Bootstrap: Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times
  - Train a classifier, C_i on D_i.



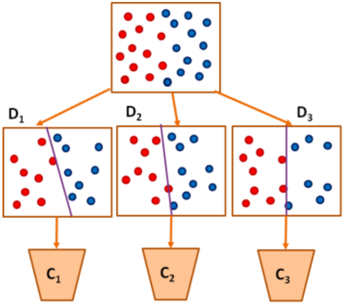
Bagging: Bootstrap Aggregation

- Bootstrap: Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times
  - Train a classifier, C_i on D_i.



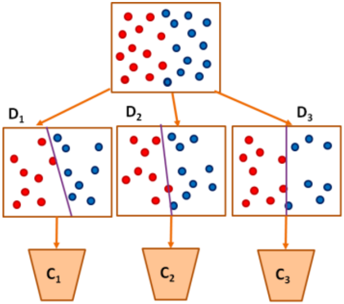
Bagging: Bootstrap Aggregation

- Bootstrap: Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times
  - Train a classifier, C_i on D_i.
- Each test sample is classified by k classifiers



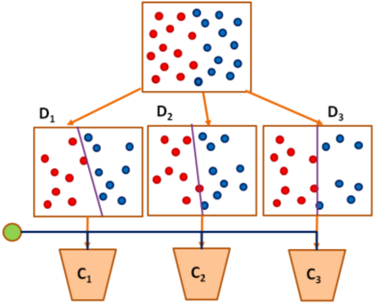
Bagging: Bootstrap Aggregation

- Bootstrap: Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times
  - Train a classifier, C_i on D_i.
- Each test sample is classified by k classifiers
- Results are averaged to obtain the final decision



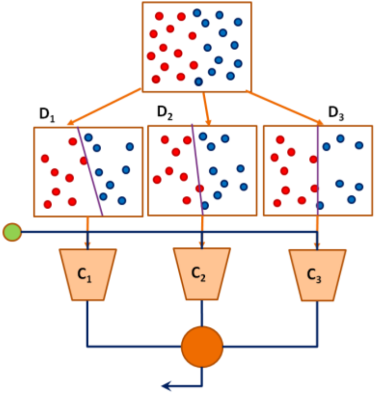
Bagging: Bootstrap Aggregation

- Bootstrap: Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times
  - Train a classifier, C_i on D_i.
- Each test sample is classified by k classifiers
- Results are averaged to obtain the final decision



Bagging: Bootstrap Aggregation

- Bootstrap: Sampling the training data with replacement (N,<N)
- Sample training data, D_i, k times
  - Train a classifier, C_i on D_i.
- Each test sample is classified by k classifiers
- Results are averaged to obtain the final decision



Bagging

- Bagging or *bootstrap aggregation* a technique for reducing the variance of an estimated prediction function.
- Bootstrap: Randomly draw datasets *with replacement* from the training data, each sample *the same size as the original training set*
- For classification, a *committee* of trees each cast a vote for the predicted class.

Random Forests

A simple way of combining Decision Trees

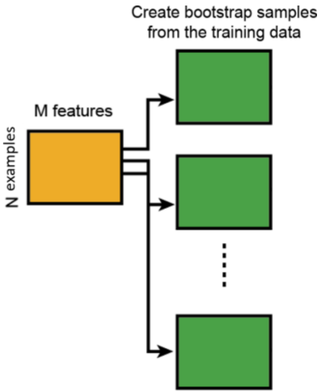
CART Algorithm: A Recap

- Classification and Regression Trees (Leo Breiman)
- Recursive Binary Splitting: Greedy Algorithm
  - All values of an attribute are sorted and all split points are tested
  - Test all such attributes and select the split with lowest cost
- Cost Functions:
  - Regression:MSE
  - Classification: Gini

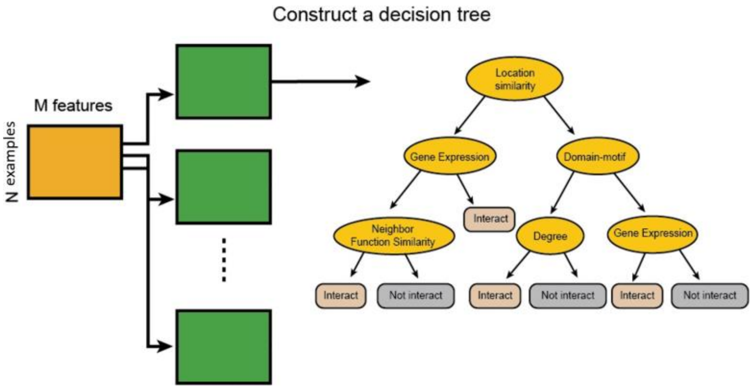
Random Forest

- **Random forest** (or **random forests**) is an ensemble classifier that consists of many decision trees and outputs the class that is the mode of the class's output by individual trees.
- The term came from **random decision forests** that was first proposed by Tin Kam Ho of Bell Labs in 1995.
- The method combines Breiman's "bagging" idea and the random selection of features.

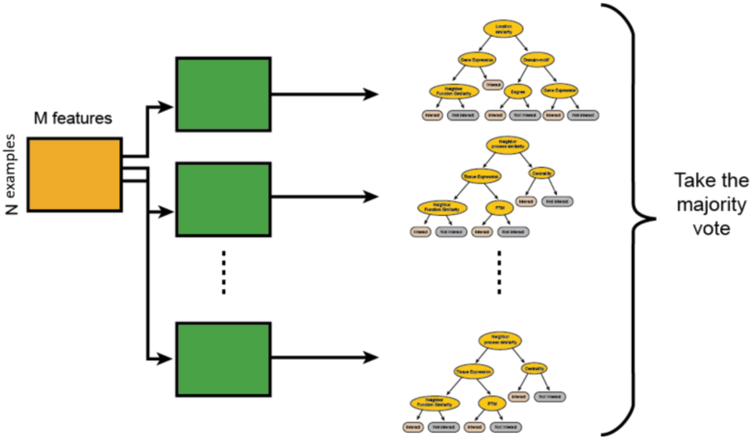
Random Forest



Random Forest

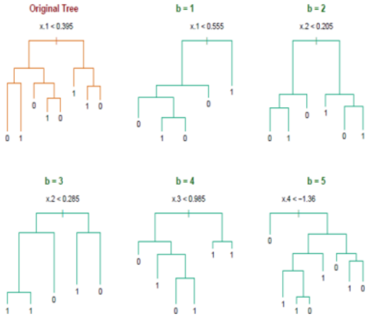


Random Forest



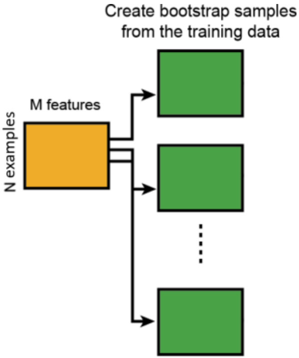
Random Forest

Random forest classifier, an extension to bagging which uses *de-correlated* trees.

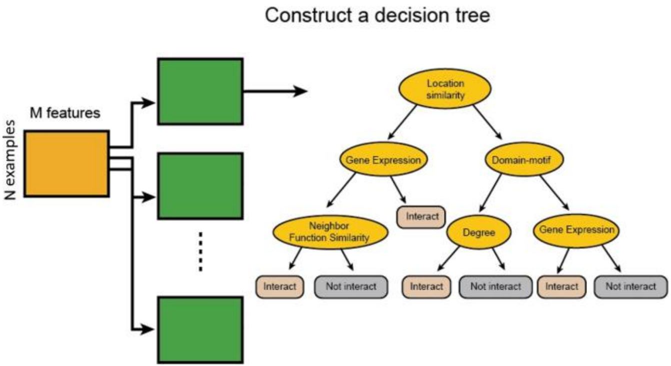


Random Forest Classifier

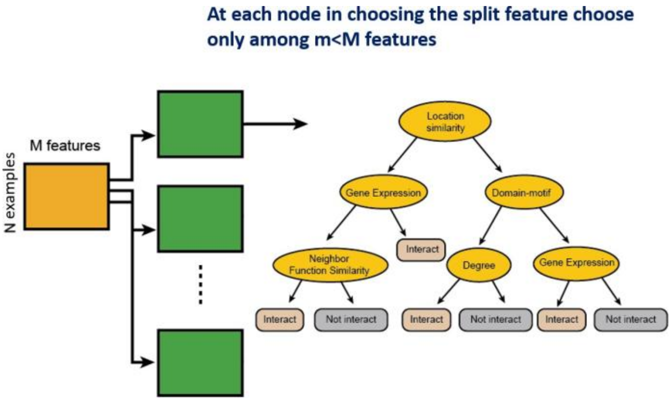
Random Forest Classifier



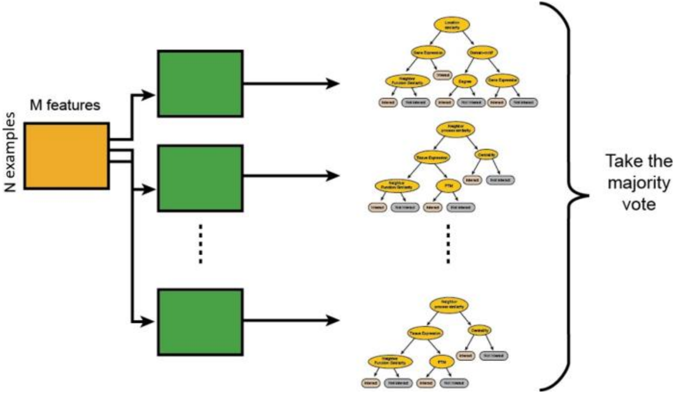
Random Forest Classifier



Random Forest Classifier



Random Forest Classifier



Example: Email Classification

Spam vs Ham

- **Features:** 2-million dimensional (one-hot)

Example: Email Classification

Spam vs Ham

- Features: 2-million dimensional (one-hot)
- Each tree selects a subset of features (words)

Example: Email Classification

Spam vs Ham

- Features: 2-million dimensional (one-hot)
- Each tree selects a subset of features (words)
    - Select the best from the subset

Example: Email Classification

Spam vs Ham

- Features: 2-million dimensional (one-hot)
- Each tree selects a subset of features (words)
  - Select the best from the subset
- Should the words be uniformly sampled?

Example: Email Classification

Spam vs Ham

- Features: 2-million dimensional (one-hot)
- Each tree selects a subset of features (words)
    - Select the best from the subset
- Should the words be uniformly sampled?
- Automatically selects relevant words

Example: Email Classification

Spam vs Ham

- Features: 2-million dimensional (one-hot)
- Each tree selects a subset of features (words)
  - Select the best from the subset
- Should the words be uniformly sampled?
- Automatically selects relevant words
- What is the equivalent in classification with Gene data

Thanks!

Questions?